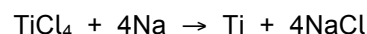




REACTING MASS CALCULATIONS 2

- 1) Titanium is extracted from titanium chloride as shown. Calculate the mass of sodium needed to react with 126 g of titanium chloride.

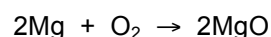


$$\text{moles TiCl}_4 = \frac{126}{190} = 0.663 \text{ mol}$$

$$\text{moles Na} = 4 \times 0.663 = 2.65 \text{ mol}$$

$$\text{mass Na} = 23 \times 2.65 = 61.0 \text{ g}$$

- 2) Calculate the mass of oxygen needed to react 9.60 g of magnesium to form magnesium oxide.

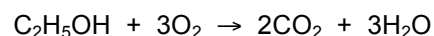


$$\text{moles Mg} = \frac{9.60}{24} = 0.400 \text{ mol}$$

$$\text{moles O}_2 = \frac{0.400}{2} = 0.200 \text{ mol}$$

$$\text{mass O}_2 = 32 \times 0.200 = 6.40 \text{ g}$$

- 3) What mass of ethanol could burn in 100 g of oxygen?

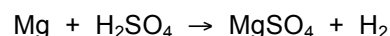


$$\text{moles O}_2 = \frac{100}{32} = 3.125 \text{ mol}$$

$$\text{moles C}_2\text{H}_5\text{OH} = \frac{3.125}{3} = 1.04 \text{ mol}$$

$$\text{mass C}_2\text{H}_5\text{OH} = 46 \times 1.04 = 47.9 \text{ g}$$

- 4) What mass of hydrogen is formed when 2.00 g of magnesium reacts with sulfuric acid?

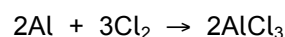


$$\text{moles Mg} = \frac{2.00}{24} = 0.0833 \text{ mol}$$

$$\text{moles H}_2 = 0.0833 \text{ mol}$$

$$\text{mass H}_2 = 2 \times 0.0833 = 0.167 \text{ g}$$

- 5) What mass of aluminium reacts with 50.0 g of chlorine to form aluminium chloride?

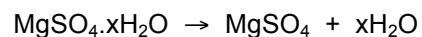


$$\text{moles Cl}_2 = \frac{50.0}{71} = 0.704 \text{ mol}$$

$$\text{moles Al} = 0.704 \times \frac{2}{3} = 0.469 \text{ mol}$$

$$\text{mass Al} = 27 \times 0.469 = 12.7 \text{ g}$$

- 6) 6.15 g of hydrated magnesium sulfate decompose to form 3.00 g of anhydrous magnesium sulfate on heating. Calculate the formula mass of hydrated magnesium sulfate and the value of x.



$$\text{moles MgSO}_4 = \frac{3.00}{120} = 0.0250 \text{ mol}$$

$$\text{mass H}_2\text{O} = 6.15 - 3.00 = 3.15 \text{ g}$$

$$\text{moles H}_2\text{O} = \frac{3.15}{18} = 0.175 \text{ mol}$$

$$\text{Ratio of moles MgSO}_4 : \text{H}_2\text{O} = 0.025 : 0.175 = \frac{0.025}{0.025} : \frac{0.175}{0.025} = 1 : 7$$

$$\therefore x = 7 \text{ (nearest whole number)}$$

Area	Strength	To develop	Area	Strength	To develop	Area	Strength	To develop
Done with care and thoroughness			Can find moles from mass			Gives units when appropriate		
Shows suitable working			Can use reacting ratios in equations			Suitable rounding		
Can work out M_r			Can find mass from moles					
Which numbers are in formula			Can find water of crystallisation					